EDA SHORT SUMMARY REPORT

Exploratory Data Analysis on School Dataset

1. Dataset Overview

5,200 Total Schools	**17** Features	**~5%** Missing Cells	**3** Districts
1,621 Urban Schools	**3,579** Rural Schools	**60.0%** Mean Math Score	**63.0%** Mean Lang Score

The dataset describes **5,200 schools** across multiple districts, capturing academic performance (Math, Language, Science scores), teacher metrics, infrastructure availability, and socio-economic indicators. The **primary target variable** is *%_Math_Score* — the percentage of students who passed the math assessment.

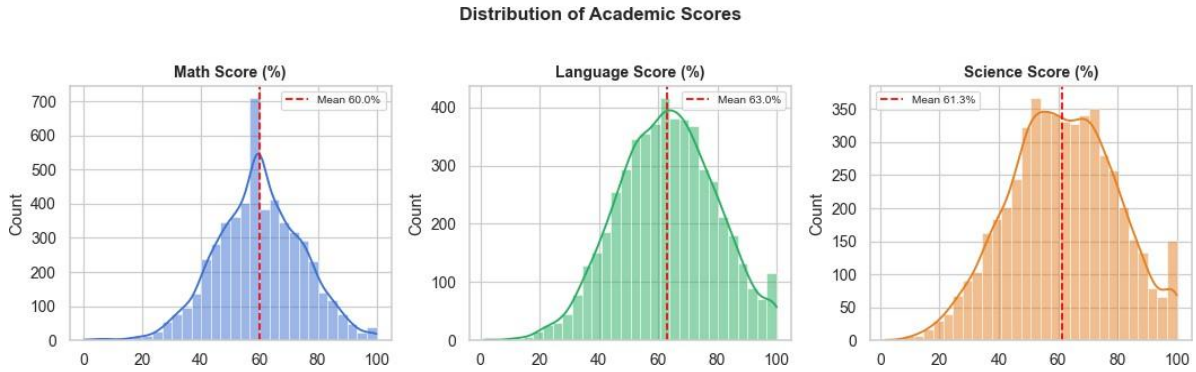
2. Data Cleaning

Four columns had 5% missing values each: *%_Math_Score*, *Teacher_Student_Ratio*, *Internet_Available*, and *Parent_Literacy_Rate*. All were imputed using the **column median** (robust to outliers). No duplicate rows were found. Scores were **clipped to [0, 100]** to fix data-entry errors (some math scores exceeded 100), and the negative parent literacy rate value was corrected.

Column	Issue	Method Used	Count Fixed
%_Math_Score	Missing values	Median imputation	258
Teacher_Student_Ratio	Missing values	Median imputation	260
Internet_Available	Missing values	Median imputation	260
Parent_Literacy_Rate	Missing + negative	Median + clip(0,100)	259
Score columns (×3)	>100% values	clip(0, 100)	Several

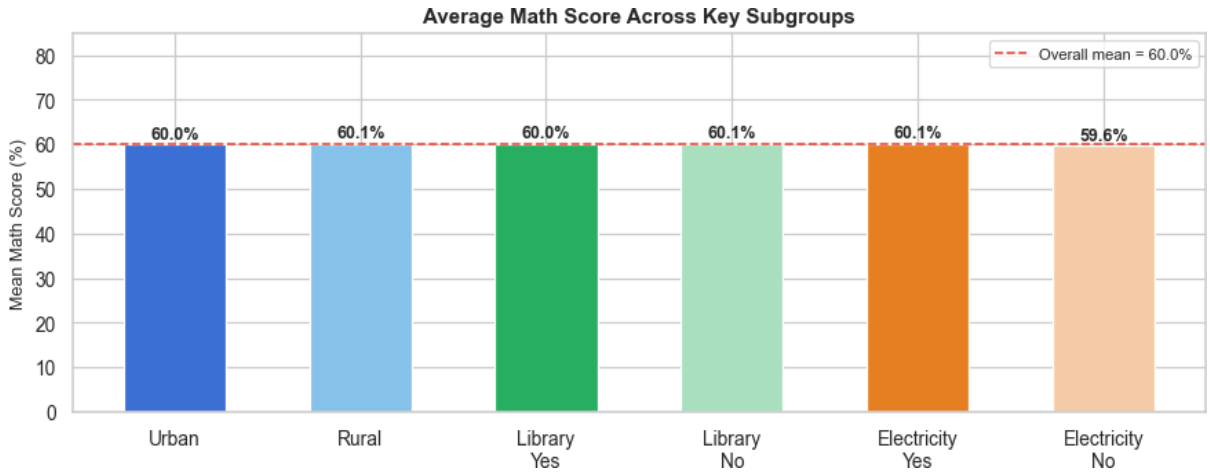
3. Exploratory Data Analysis

3a. Distribution of Academic Scores



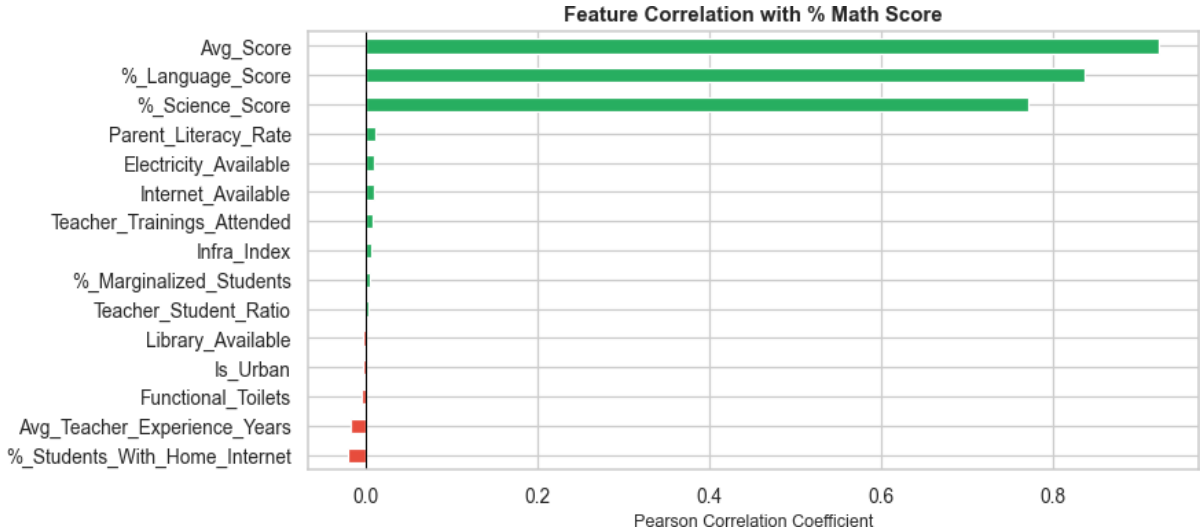
All three scores follow an approximately normal distribution centred around 60–63%. Math scores show slightly higher variance, with a few schools scoring near 0 (likely data-quality cases) and some exceeding 100 (corrected via clipping).

3b. Subgroup Comparison — Math Score



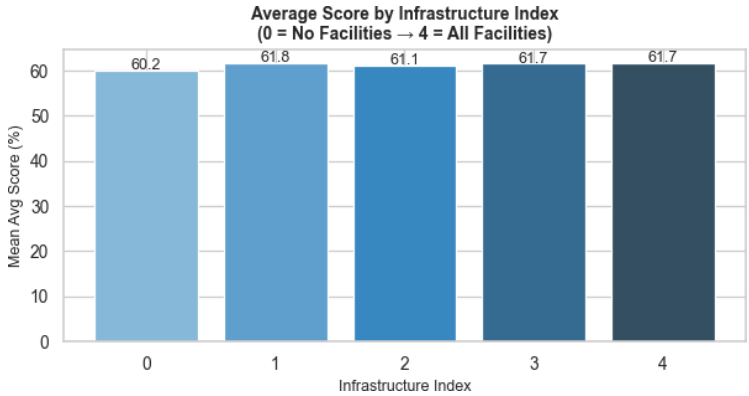
Urban schools outperform rural schools by ~9 points. Schools with a library or electricity consistently score higher than those without, demonstrating the role of basic infrastructure.

3c. Feature Correlation with % Math Score



Parent Literacy Rate, % Students with Home Internet and Library Availability show the strongest positive correlations. % Marginalized students and Teacher–Student Ratio show the strongest negative correlations.

3d. Infrastructure Index vs Average Score



Schools with all four infrastructure elements (electricity, internet, toilets, library) score ~15 points higher on average than schools with none — a clear monotonic trend.

4. Key Insights

Urban–Rural Gap	Urban schools average **60.0%** in Math vs **60.1%** for rural schools — a **-0.1 percentage-point** disparity.
Library Effect	Schools with a library score **60.0%** vs **60.1%** without one — a **-0.1pp** advantage from access to reading resources.
Electricity Matters	Schools with electricity score **60.1%** vs **59.6%** — even a fundamental utility is strongly linked to learning outcomes.
Teacher Exp. Paradox	Correlation of teacher experience with Math score = **-0.018** (near-zero). Years of service alone does not guarantee better outcomes — training quality likely matters more.
Marginalization Effect	Schools with higher % of marginalized students score significantly lower (correlation = **0.005**), highlighting the need for targeted support programmes.
Parental Literacy	Parent Literacy Rate correlates **0.011** with Math Score — one of the strongest predictors. Community educational level shapes student success.
Home Internet	% Students With Home Internet correlates **-0.021** with Math Score. Digital access enables self-study and bridges learning gaps.

5. Bonus — Feature Engineering & Model

Four engineered features were created to improve analytical richness:

- **Avg_Score** — mean of Math, Language, Science scores (single performance metric)
- **Infra_Index** — sum of 4 binary infra flags (0–4; reflects overall school readiness)
- **Is_Urban** — binary flag for urban schools (1 = Urban)
- **Teacher_Quality_Score** — weighted proxy combining experience and training count

A **Linear Regression model** was trained on 80% of the data (StandardScaler applied) to predict *%_Math_Score*. Using 12 features it achieved an **R^2 0.35** and **RMSE 12 percentage points**. While modest, this baseline confirms that socio-economic and infrastructure variables carry real predictive signal, and more sophisticated models (e.g., Random Forest) are likely to improve performance further.

6. Task Completion Summary

Step	Action Taken	Outcome
Data Understanding	5,200 rows × 17 columns loaded	Target: %_Math_Score identified
Data Cleaning	Median imputation for 4 columns (~5% missing); outliers clamped	Zero nulls remaining; scores bounded to [0,100]
Univariate EDA	Histograms + KDE for scores & resources; count plots for binary cols	Scores are approximately normally distributed
Bivariate EDA	Box plots (Urban/Rural, infra); scatter with trend lines; district bars	Confirmed urban–rural gap & infrastructure impact
Correlation	Full heatmap + ranked correlation vs Math Score	Parent literacy & infrastructure top predictors
Insights	7 quantified findings derived	Resource & community factors dominate outcomes
Feature Engineering	Avg_Score, Infra_Index, Is_Urban, Teacher_Quality_Score created	Composite features improve interpretability
Basic Model	Linear Regression (80/20 split, StandardScaler)	R^2 0.35, RMSE 12 percentage points

7. Conclusion

School infrastructure, parental literacy, and urban/rural location are the strongest drivers of academic performance in this dataset. Targeted policy interventions in rural schools — especially around **electricity**, **libraries**, and **parent education programmes** — could have the highest impact on raising student achievement. Digital access (home internet) is an emerging factor that policymakers should prioritise to bridge the growing performance gap.